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Labour market institutions: an obstacle or support to Latvian labour market recovery?

Anna Zasova^{1,2}

1. Introduction

In 2008, the Latvian economy experienced the strongest shock in the post-transition period. The repercussions of the shock have extended to all sectors of the economy and have impacted essentially all socio-economic groups of the population. The recovery took place under an unchanged currency peg and in circumstances where the government's capacity to use fiscal stimulus was very limited, which left most burden of adjustment on the ability of labour and product markets to react flexibly. In a sense, in light of Latvia's prospective eurozone membership, the recovery from the crisis provides a very useful insight into the effectiveness of adjustment mechanisms which will be available when the exchange rate instrument becomes unattainable.

In late 2009, output growth resumed and continued throughout 2010. However, it considerably lags behind the pace of recovery in the neighbouring Baltic states, despite Latvia having experienced the deepest output contraction in the aftermath of the crisis ("peak-to-trough" fall in GDP amounted to 25.5% in Latvia, 20.1% in Estonia and 17.0% in Lithuania, whereas cumulative growth since the beginning of recovery has been 3.8%, 10.3% and 8.2%, respectively³). The unemployment rate in the Baltics increased significantly (in Latvia - from the lowest level of 5.3% to 20.5%, in Estonia - from 4.1% to 19.8%, in Lithuania - from 4.5% to 18.3%⁴) and, despite a reduction in the latest quarters, remains high. What are the determinants of labour market adjustment to long-term equilibrium and equilibrium itself? What can one say about the potential of the Latvian economy to return to a single digit unemployment rate in the foreseeable future? A vast amount of academic literature suggests that the speed of labour market adjustment to shocks and the equilibrium in the labour market depend on labour market institutions. For example, institutions may have an impact on employers' incentives to hire new employees if uncertainty exists about future demand, or incentives to hire low qualified and thus relatively less productive workers. Also, people's incentives to engage in active job search instead of relying on generous unemployment or other benefits and their reservation wages are shaped by labour market institutions.

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³ Source: author's calculations based on Eurostat data. GDP growth in Estonia in the 1st quarter of 2011 is a flash estimate of Statistics Estonia

⁴ Source: Eurostat

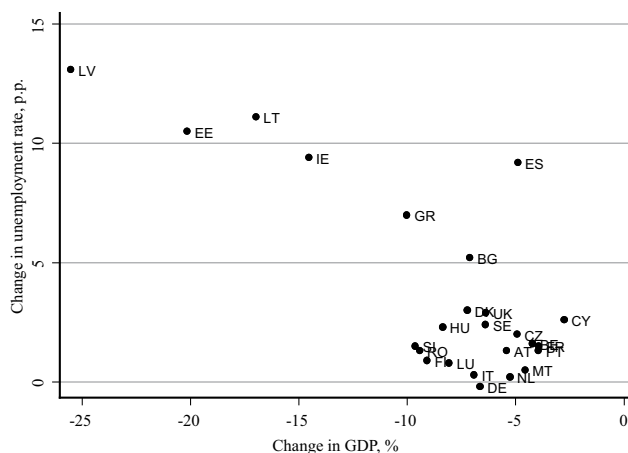
This article will assess Latvian labour market institutions in a market flexibility perspective, by making comparisons with other European countries, especially focusing on the other Baltic states. I will analyse the policy changes that have been implemented since the outbreak of the crisis, attempting to evaluate whether the changes are likely to make the market more flexible or, on the contrary, are likely to increase distortions in the market. I will consider the likely impact of, and changes, if any, in the following labour market institutions: the minimum wage, employment protection legislation, and tax burden on labour. I will argue that the seemingly flexible reaction of the Latvian labour market to the crisis is to a large extent due to weak law enforcement, whereas formal regulations are rather rigid and are not much different from the EU-15.

The paper is organised as follows: the following section presents a brief overview of labour market developments since 2008, focusing on indicators that characterise the ability of labour markets to adjust to shocks. The third section presents an analysis of the institutions and their changes since the beginning of recession, the fourth section concludes.

2. Labour market adjustment during the crisis

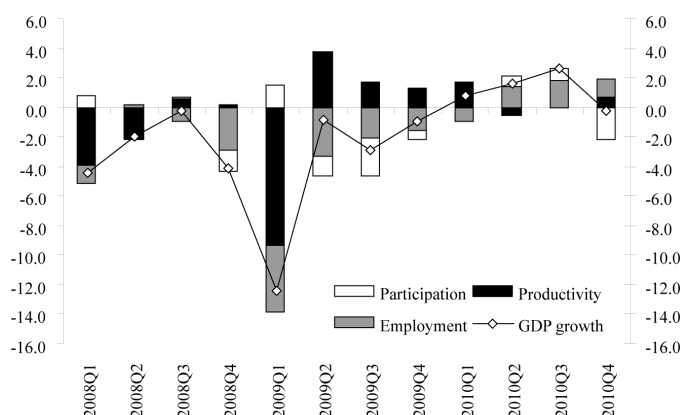
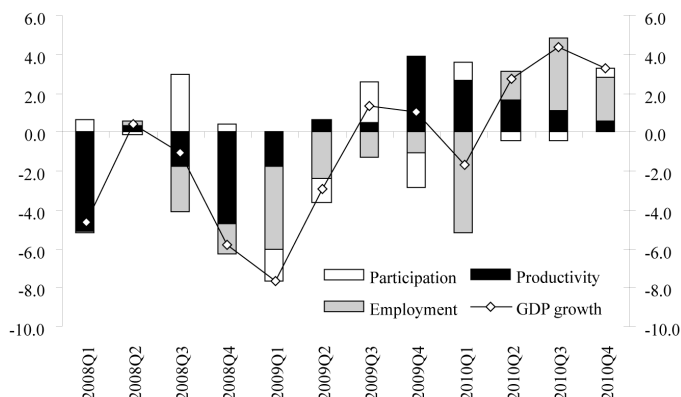
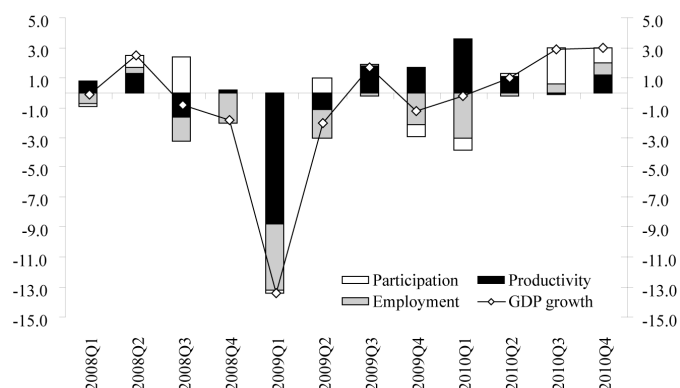
The economic slowdown in the Baltics, which began even before the bankruptcy of Lehman Brothers in September 2008, being initially triggered by the end of a domestic demand boom, and then reinforced by external conditions, turned into a deep recession, which resulted in severe output contraction in all three countries. Latvia, which was the most overheated economy before the crisis, also experienced the strongest GDP fall, which in the period from the beginning of 2008 to the 3rd quarter of 2009 accrued to more than 25%. The fall in GDP was accompanied by a dramatic increase in the unemployment rate, which in Latvia was by far the strongest among the EU economies (see Figure 1).

Figure 1: Real GDP contraction from peak to trough (%) and change in unemployment rate in the respective period (percentage points) in EU-27⁵



Source: Eurostat, author's calculations

⁵ Countries that did not have 2 consecutive quarters of negative growth – Poland and Slovakia – are not included.

Figure 2: Contribution of changes in labour participation, employment and labour productivity to GDP (per person of working age) growth in the Baltic states in 2008-2010**Latvia****Estonia****Lithuania**

Source: Eurostat, author's calculations

Looking from the production side, growing unemployment and falling labour productivity were the main driving forces behind the GDP fall in the initial phase of the crisis in Latvia and the other Baltics, whereas the labour participation rate declined relatively less (in Latvia and Estonia) or even increased (in Lithuania). GDP dynamics can be decomposed into the above three elements in the following way:

$$\frac{GDP}{POP} = \frac{GDP}{N} * \frac{N}{L} * \frac{L}{POP} \quad [1]$$

Where the left hand side of the equation is real GDP per person of working age (POP^6); N is total employment, thus the first term on the right hand side is real GDP per person employed, describing labour productivity; L is the economically active population, thus the second term is 1 minus the unemployment rate; and the last term is the labour participation rate. Figure 2 presents decomposition of real GDP growth into the above three components in the Baltics over 2008-2010, which is based on the linearized version of equation [1].

The initial phase of the crisis in all three Baltics (up to Q1 2009) was accompanied by huge employment and labour productivity losses. In the subsequent periods labour productivity started to pick up and turned predominantly positive from the second half of 2009. Employment in Latvia and Estonia continued to decline until the 1st quarter of 2010, but in the remaining quarters of 2010 employment gains became the main driving force of GDP growth. In Lithuania, which faced smaller employment losses in the initial phase of the crisis, the negative trend persisted longer and growth in the second half of 2010 was much more modest.

How does employment dynamics in the Baltics during the recovery process compare to the rest of the EU countries? Is the GDP recovery, which in most EU countries has been underway since the second half of 2009 – early 2010 reflected in a gradually falling unemployment rate? The answer is “no” for most eurozone economies (see Figure 3): even in countries where GDP gains from the lowest recession point have been quite strong (e.g., Malta, the Netherlands, Denmark), the unemployment rate is still above the trough point. The strongest unemployment reduction since the beginning of GDP recovery was observed in Latvia and Germany. It is true that Latvia also had the biggest increase in unemployment in the period preceding recovery. However, in the countries that experienced a comparable rise in unemployment and had a similar unemployment rate at the beginning of GDP recovery – Spain and Lithuania – the unemployment rate is still above the end of recession level and in Spain the unemployment rate is still rising.

What are the reasons for diverse adjustment in the European labour markets and what are possible explanations for some countries having a seemingly more sluggish unemployment reaction to GDP recovery? Persistence of European unemployment has been at the centre of academic dispute since the 1970s-1980s, being fuelled by scholars’ attempts to explain long-term differences in unemployment dynamics in the US and Europe. A vast amount of literature suggests that the speed of labour market adjustment to equilibrium unemployment and long-term unemployment itself are determined by labour market institutions.

The impact of institutions can be generalised as being twofold. First, the institutions determine persistence and level of unemployment by affecting responsiveness of real wages and prices to the unemployment rate (Layard et al, 2005, ch. 8). For example, strict employment protection legislation increases the bargaining power of insider workers, which makes wages less responsive to the unemployment rate: it makes firing more costly, and hence, at an un-

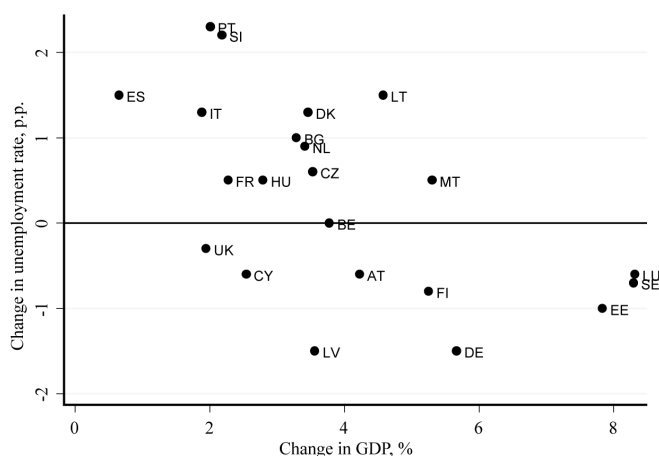
⁶ Population aged 15-74.

changed rate of unemployment, the risk of losing a job for an insider is lower and accordingly he/she can bargain for a higher wage. High minimum wages can also reduce employers' ability to manoeuvre and can reduce their incentives to hire low qualified workers, thus increasing the bargaining power of insider workers at an unchanged rate of unemployment.

The second dimension of institutional impact refers to the quality of matching between a worker and an employer. Any institution, which is likely to increase the quality of job matching is expected to reduce flows from employment to unemployment, because of better correspondence of the qualifications of the employed to the requirements of the employers. For example, if an unemployed person can spend more time on job search while receiving unemployment benefit, the probability of a successful job match is higher and hence job separation becomes less likely.

Efficiency of job match can be inferred from the relationship between the unemployment rate and availability of vacancies, the so called Beveridge curve. The vacancy rate (the ratio of unfilled vacancies to the sum of unfilled vacancies and the number of jobs filled) and the unemployment rate are expected to be negatively correlated: an increase in the vacancy rate should lead to a reduction in the rate of unemployment. However, if a mismatch occurs between the skills demanded and supplied in the market, the increase in the vacancy rate may leave the unemployment rate unaffected. Therefore, an outward shift in a Beveridge curve is a signal of worsening efficiency of job match. Beveridge curves for the Baltics suggest that the gradual rise in the vacancy rate which was observed during the recovery was accompanied by a reduction in the rate of unemployment (see Figure 4). Latvia and Lithuania seem to move along the curves, whereas in Estonia there was a slight outward shift of the curve in

Figure 3: Real GDP recovery from trough to 4Q 2010 (%) and change in unemployment rate during the respective period (percentage points) in EU-27⁷

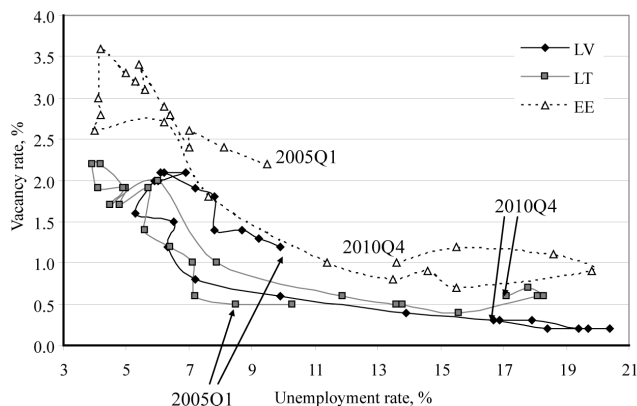


Source: Eurostat, author's calculations

⁷ Countries that did not have 2 consecutive quarters of positive growth by 4Q 2010 (Ireland, Greece and Romania), as well as countries that did not experience 2 consecutive quarters of negative growth (Poland and Slovakia) are not included.

the initial stage of the recovery, but in recent quarters the decline in unemployment was not accompanied by a growing vacancy rate, suggesting a gradual improvement in the job match.

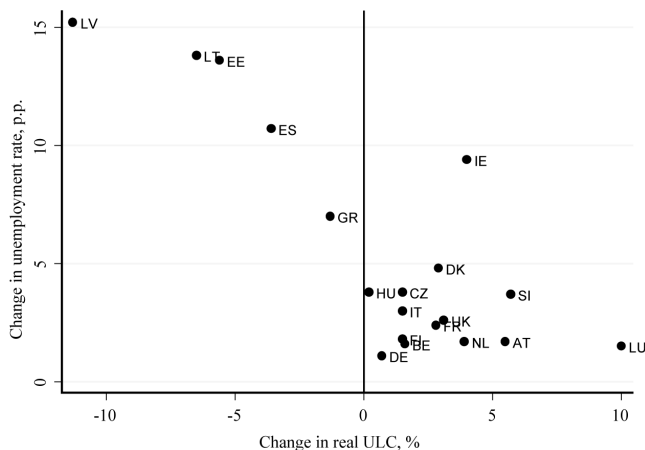
Figure 4: Beveridge curves for the Baltic states, 2005Q1 – 2010Q4



Source: Eurostat

Finally, labour market flexibility can also be discussed in terms of wage adjustment. Wage reduction in response to the increase in the unemployment rate in Latvia and also in the other Baltic States was substantial during the recession (see Figure 5). Real unit labour costs (ULC) in Latvia fell by more than 10%, and by 5-6% in Estonia and Lithuania, which is by far the largest reduction across the EU member states. In fact, real ULC in most EU countries did not decline while the unemployment rate was rising.

Figure 5: Adjustment in real ULC (%) and unemployment rate (percentage points) during the recession in EU-27 countries⁸



Source: Eurostat, author's calculations

⁸ In Figure 5, the beginning of the adjustment period is defined as the start of unemployment growth after the beginning of the recession (i.e., after two consecutive quarters of negative GDP growth), but the end of the adjustment period is defined as two consecutive quarters of unemployment reduction

Thus, both efficiency of job match and wage response to the recession suggest that labour market adjustment in Latvia has so far been effective. The question is – what is the potential of the Latvian economy to return to a low unemployment environment in the foreseeable future and are the labour market institutions supportive of recovery?

3. Labour market institutions in Latvia vs. other EU countries

3.1 Minimum wage

The impact of a statutory minimum wage on employment and overall functioning of the labour market is not straightforward: on the one hand, plenty of empirical evidence suggests that a minimum wage reduces employment (for a review of empirical literature on the impact of minimum wage, see Neumark and Wascher, (2006))⁹, especially among low-skilled and thus less productive workers. On the other hand, a minimum wage reduces poverty and increases the income of those who continue working and, thus, the ultimate judgement about the impact of a minimum wage may depend on the priorities of the state.

A minimum wage is set by statute in 18 out of 27 EU countries (Eurostat, 2011).¹⁰ Minimum wages in Belgium and Greece are determined by collective bargaining, but since the minimum wage coverage in these countries is very broad, these countries are also included in the analysis in this section. In order to compare the impact of minimum wages in these countries, I will analyse the ratio of the minimum wage to average compensation of employees from the data on national accounts. A common approach in the literature when it comes to making cross-country minimum wage comparisons is to consider the relative level of minimum wage relative to the average or median wage. Relating the minimum wage to the median wage has certain advantages compared to using the average wage or average compensation of employees, since the former gives a better idea of how many people could be affected by an increase in the minimum wage, taking into account that wage distribution is not symmetrical (e.g., if the share of low wage earners is large, but the number of people receiving very high wages is very small, a minimum wage ratio to the average wage or average compensation of employees will underestimate the potential impact of an increase in the minimum wage).

Unfortunately, statistics on the median wage in Latvia do not exist, whereas average wage statistics have an important weakness compared to average compensation of employees from the national accounts. National account data is adjusted for the statistical office's estimate of the shadow economy. Therefore, compensation of employees from the national accounts at least partially takes account of so-called "envelope wages". This is crucial especially in the case of Latvia, since, according to available evidence, the prevalence of envelope wages in Latvia is among the highest in the EU. As shown by Williams (Williams, 2009) 17% of those

⁹ Although empirical evidence on the impact of the minimum wage is rather mixed, providing a very broad interval of estimates, Neumark and Wascher (2006) argue that on the whole the empirical evidence is consistent with theoretical predictions that the minimum wage creates distortions in the market and adversely affects employment, at least in the segment of least skilled workers.

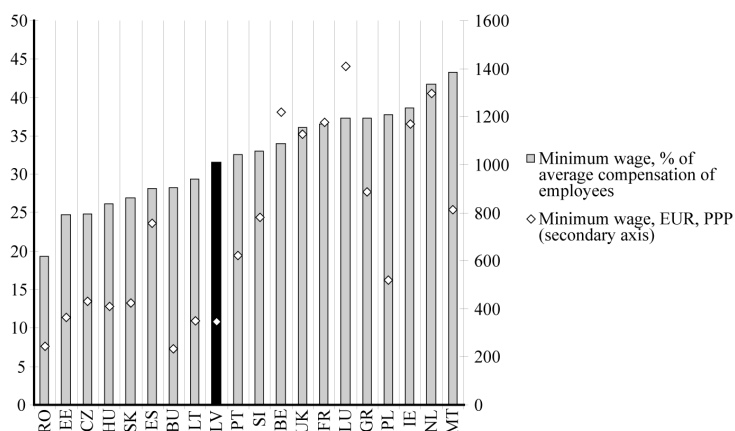
¹⁰ A statutory minimum wage exists in Bulgaria, the Czech Republic, Estonia, France, Hungary, Ireland, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain and the United Kingdom.

formally employed in Latvia (versus 5% in the EU-27, 11% in Lithuania and 8% in Estonia) receive envelope wages amounting on average to 46% of their total wage income. The only EU country where the share of envelope wage receivers is estimated to be higher is Romania (23%). Since data on average wages come from enterprises' administrative reports, it obviously does not account for envelope wages and, therefore, making cross-country comparisons by relating the minimum wage to the average wage can overestimate the sensitivity of the Latvian labour market to the minimum wage. Of course, if a person is employed illegally, the minimum wage is not binding; therefore, the ratio of the minimum wage to the average compensation of employees characterises the sensitivity of the labour market to the minimum wage only for those employed officially. Figure 6 presents the minimum wage ratio to the average compensation of employees and the minimum wage level in euro at purchasing power parity (PPP) in EU countries.

The minimum wage level in euro at PPP in Latvia, as well as the other Baltic states, is among the lowest in the EU. However, if one compares the minimum wage ratio to average compensation of employees, in Latvia it is, as seen from Figure 6, one of the highest among the new EU member states (31.6% in 2010), higher than in Estonia (24.7%) and Lithuania (29.3%), which suggests that the impact of the minimum wage on the labour market in Latvia is likely to be relatively strong. The fact that the minimum wage *level* in Latvia is relatively low, but its *ratio* to average compensation of employees is high, can be explained by the fact that labour in Latvia is relatively less productive and hence, earns less.

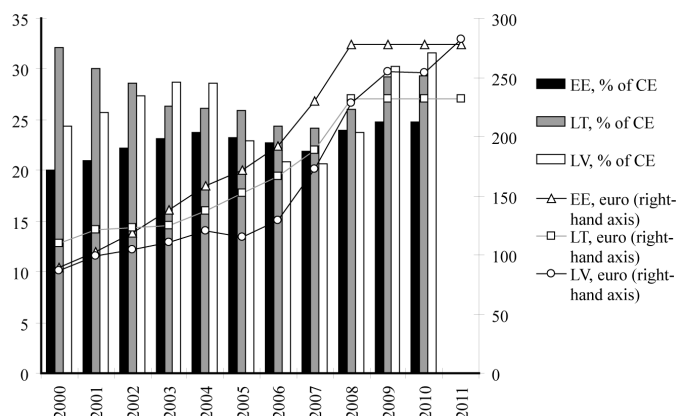
Latvia is the only Baltic state to have raised the minimum wage in the post-crisis period (see Figure 7). Neither Estonia nor Lithuania have amended their minimum wage since 2008, whereas in Latvia the minimum wage has been raised twice – on 1 January 2009 (from 160 LVL (228 EUR) to 180 LVL (256 EUR) per month) and on 1 January 2011 (to 200 LVL (285 EUR) per month) and, as a result, the minimum wage in Latvia is higher than in the other Baltics both relative to average compensation of employees and after the last increase – in euro terms.

Figure 6: Minimum wage ratio to average compensation of employees (%) and minimum wage at PPP (euro) in EU countries in 2010



Source: Eurostat, author's calculations

Figure 7: Minimum wage level in the Baltics in 2000-2011 (euro, left-hand axis) and its ratio to average compensation of employees (CE) (% , right-hand axis)



Source: Eurostat, author's calculations

In 2003, the Latvian government passed regulations on minimum wage determination, which foresaw raising the minimum wage to 50% of the previous year's average gross monthly wage in the economy by 2010. However, the target was not achieved (in 2010 the minimum wage amounted to 39.0% of the gross monthly wage in 2009; in 2011, the minimum wage was raised to 44.9% of the average wage in 2010) and in 2011 the Cabinet of Ministers passed a new policy paper on determining the minimum wage in the following years (Ministry of Welfare of Latvia, 2011). The policy paper foresees raising the minimum wage to the level of 47%-48% of the average gross monthly wage by 2014. The policy paper also argues that in order for Latvia to be able to ratify the 4th article of the European Social Charter (which makes provision for fair labour remuneration (Council of Europe, 1961)), the ultimate goal should be to raise the minimum wage to 68% of the average wage.

None of the EU member states is currently even close to having a minimum wage which constitutes 68% of the average wage. It is true that the data on average wages, which is used by policy makers when deciding on the minimum wage level, suffers from many drawbacks and does not allow inferring the actual number of people that might be affected by an increase in the minimum wage with any acceptable degree of certainty. However, the conjecture is that the Latvian minimum wage relative to labour earnings is already rather high as compared to other new member states, which means that further increase of the minimum wage relative to average earnings is likely to increase relative distortions. After all, it has to be remembered that the expected positive impact on the welfare of the employed is only one side of the coin, while the other side of the coin are increased disincentives for employers to hire low-skilled workers and increased motivation to increase informal employment.

3.2 Employment protection legislation (EPL)

Stringent EPL norms are expected to reduce flows in the labour market and to constrain labour market adjustment to a recession by restricting employers' ability to dismiss workers in

response to a worsening economic environment. This does not necessarily imply, however, that strict EPL should result in higher employment, since inability to adjust employment during economic downturns can also create disincentives for employers to hire people during upturns (European Commission, 2007). Strict EPL can also impede labour market adjustment by reducing wage responsiveness to unemployment, as in case of strict employment protection rules insider workers enjoy higher bargaining power.

I will analyse EPL strictness in Latvia vs. other EU countries, using three alternative indicators: (i) the OECD EPL index (Venn, 2009), (ii) the World Bank rigidity of employment index (The International Bank for Reconstruction and Development / The World Bank, (2010)) and (iii) the World Economic Forum Hiring and Firing Practices index (World Economic Forum, 2010.).

The OECD EPL index (Venn, 2009) is probably the most widely used index in employment protection analysis (see e.g. Bassanini and Duval, 2006, Boeri et al, 2004, Eamets, Masso, 2004, Bertola et al, 1999). It aggregates 21 norms, which characterise three dimensions of legislation: (1) individual dismissals of workers with regular contracts, (2) temporary employment and (3) additional regulations for collective dismissals. Each legislative norm is assigned a score ranging from 0 (fully flexible) to 6 (fully rigid), but the aggregate index is calculated as a weighted average of the scores (for a detailed description of methodology, see Venn (2009)). The EPL indices are calculated for all OECD member states and selected candidate countries (latest available data refer to 2008), but, since Latvia is neither a member state, nor a candidate country, the OECD does not produce an EPL index for Latvia. However, in his recent paper, Muravyov (2010) assessed evolution of employment protection legislation norms in the CIS and Baltic countries over the period from 1985 to 2009, using OECD methodology and I shall use this paper as a source of data on Latvia (and Lithuania for comparative analysis).

The World Bank rigidity of employment index is calculated within the *Doing Business* study of entrepreneurial environment. In the 2009 year study (The International Bank for Reconstruction and Development t/T The World Bank, 2009) assessment was based on 10 criteria, one of which being the rigidity of employment index, which evaluates three aspects of employment legislation: (1) difficulty of hiring, (2) rigidity of hours and (3) difficulty of redundancy. On each of the above positions, countries are assessed on a scale of 0 (fully flexible) to 100 (fully rigid) and an aggregated rigidity of employment index is calculated. The latest available data on the World Bank rigidity of employment index refers to 2009, since in 2010 the employment rigidity index was temporarily excluded from the study due to changes in methodology.

The World Economic Forum Hiring and Firing Practices index forms part of the Global Competitiveness Report, where it is used to assess countries' labour market efficiency (for a detailed description of methodology see World Economic Forum (2010)). The Hiring and Firing Practices assessment is based on a survey of enterprises, not on formal regulations. Therefore, the World Economic Forum index represents an alternative data source on flexibility of hiring and firing practices, since both the OECD and the World Bank indices are based on formal legislation. The index is based on opinions of business executives, who are asked to characterize hiring and firing of workers on a scale ranging from 1 (impeded by regulations) to 7 (flexibly determined by employers).

Table 1 shows the values of the above indices for Latvia, Lithuania, Estonia, average value in the Baltic states¹¹ and EU-15.

Table 1: Employment protection legislation: strictness in Latvia, Lithuania, Estonia and EU-15

	Latvia	Lithuania	Estonia	Baltic States	EU-15
OECD EPL index, 2008 (0 – fully flexible, 6 – fully rigid)	2.39	2.61	2.27	2.46	2.34
The World Bank rigidity of employment index, 2009 (0 – fully flexible, 100 – fully rigid), of which	43	38	51	44	35
Difficulty of hiring	50	33	33	39	32
Rigidity of hours	40	60	60	53	40
Difficulty of redundancy	40	20	60	40	32
The World Economic Forum Hiring and Firing Practices Index, 2009 (1 – fully rigid, 7 – fully flexible)	4.6	4.6	4.9	4.7	4.6

Source: Muravyov, (2010); The International Bank for Reconstruction and Development / The World Bank, (2009); World Economic Forum (2010); Venn (2009); author's calculations

The OECD and the World Bank indices suggest that EPL in the Baltics on average is more stringent than in the EU-15, whereas the World Economic Forum index implies that Baltic EPL is more flexible. The World Bank index suggests that Lithuanian EPL is least stringent, whereas the OECD and World Economic Forum indices are most flexible for Estonia.

It is noteworthy that the average Baltic EPL indices, which are based on compilation of legislative norms (i.e., OECD and World Bank indices) suggest relatively more stringent EPL as compared with the EU-15, whereas the World Economic Forum index, which is based on subjective employers' evaluation, implies the opposite, which might be an indication of relatively weak law enforcement in the Baltics. In order to test this more formally, I shall use the method similar to that applied by Eamets and Masso (2004): the authors estimated the relationship between the OECD EPL index and the IMD World Competitiveness Yearbook index of labour regulations, the latter being based on an opinion survey of business executives for two groups of countries – “old” EU member states and a group of five Central and Eastern European (CEE) countries. By showing that the estimated relationship is statistically significant for the old EU member states but not significant for the CEE economies, the authors argue that this might be an indication of poorer law enforcement in CEE economies.

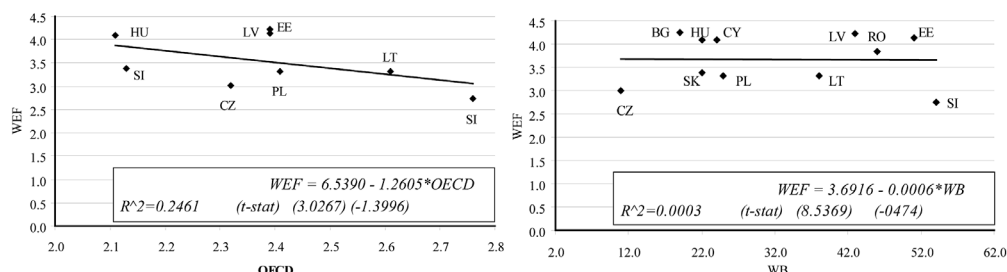
In this paper, I estimate the relationship between two pairs of indices – the OECD EPL index (OECD) vs. the World Economic Forum index (WEF) and the World Bank index (WB) vs. the WEF index. Current data availability allows testing whether Eamets' and Masso's (2004) results hold on more recent data and also allows inclusion of additional countries in the group of “new” member states. In this exercise countries are divided in two groups – EU-12 and EU-15. Figure 8 presents the results.

The World Bank, (2009); World Economic Forum (2010); Venn (2009); author's calculations
According to the results, the WB and OECD are not significant determinants of employers' perception of hiring and firing rigidity in EU-12 countries, whereas in the EU-15 the relationship is statistically significant (the estimated coefficient is negative, since the WEF index

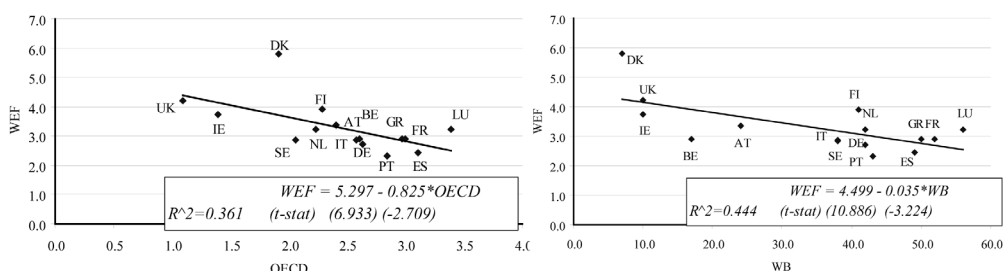
¹¹ Simple average, calculated by the author.

Figure 8: World Economic Forum Hiring and Firing Practices index (WEF, 2008) vs. OECD EPL index (2008) and World Bank Rigidity of Employment index (WB, 2009) in the EU-12 and EU-15¹²

EU-12



EU-15



Source: Muravyov, (2010); The International Bank for Reconstruction and Development

increases with flexibility, while the higher values of WB and OECD mean more rigid employment protection practices). These findings confirm the earlier results of Eamets and Masso (2004), who based their analysis on 1999 and 2001 data, and can be interpreted as a signal that enforcement of employment protection legislation in the EU-12 remains weak relative to the EU-15. The WEF index for Latvia is one of the highest among the EU-12, suggesting that employers perceive hiring and firing practices as being relatively flexible, while the WB and OECD indices, on the contrary, imply relatively rigid employment protection legislation. This discrepancy might suggest that the situation with labour law enforcement in Latvia is one of the worst even among the EU-12.

Despite the fact that in recent years Latvia has passed several important legislative norms aimed at strengthening labour law enforcement (e.g., in 2008, the fine on employers for not having a written contract with employees was raised several times), there is still no clear evidence of a notable improvement. Table 2 reports the number of enterprises inspected by the State Labour Inspection (SLI) and the number of cases of illegal employment revealed:

The data suggest that the number of cases of illegal employment revealed declined both in 2008 and 2009. However, according to SLI information, the reduction was due to certain improvements only in the construction sector, whereas in other problem sectors (woodworking,

¹² The World Bank study does not cover Malta. Data on the OECD ELP index is not available for Bulgaria, Cyprus and Romania.

Table 2: Number of enterprises inspected by State Labour Inspection and number of cases of illegal employment revealed in 2005 - 2009

	2005	2006	2007	2008	2009
Number of enterprises inspected	1994	3718	3987	4554	4996
Number of cases of illegal employment revealed	936	1802	2846	1623	1211

Source: State Labour Inspection (2010), State Labour Inspection (2009)

food production) the number of cases detected increased, which was due to the crisis-driven increase in competition in the labour market (State Labour Inspection, 2009). The improvement in the situation in the construction sector is likely to be partly due to a very strong output fall in this sector, but was also due to certain improvements in legislation¹³.

Another reason for the declining number of detected illegal employment cases is that detection becomes more complicated as entrepreneurs invent new ways of circumventing the rules (State Labour Inspection, 2010). For example, in 2009 the number of detected so-called “first day contracts” (which means that, according to the date indicated in the contract, the contract is signed one day before the inspection) increased significantly. However, in early 2010 amendments to the Cabinet of Ministers “Regulations Regarding Registration of Persons Making Mandatory Payments of State Social Insurance and Reports Regarding Mandatory Payments of State Social Insurance and Personal Income Tax” were passed. These stipulate that an employer has to notify the State Revenue Service at least one day before hiring a new worker, which is likely to have reduced the prevalence of “first day contracts”.¹⁴

There is additional evidence of an increased number of violations of labour laws in the post-crisis period. According to SLI data, the number of officially registered unemployed among detected illegally employed persons increased significantly: in 2009, 8-10% of all illegally employed persons were registered as unemployed, whereas in 2008 the corresponding amount was only 2-3%. It is quite common that workers deliberately choose to interrupt official work relationships to register as unemployed and simultaneously receive unemployment benefits (State Labour Inspection, 2010).

To conclude, the strictness of employment protection legislation in Latvia and the other Baltics is not much different from the EU-15, which, compared to other developed countries outside the EU, is characterised as being rather restrictive. At the same time, there is evidence suggesting that law enforcement in the EU-12 is significantly weaker than in the EU-15, which leaves employers room for manoeuvre and allows for a relatively flexible response to a changing economic environment. However, with the ongoing perfection of legislation and potential strengthening of state institutions' administrative capacity, one may expect a gradual improvement in law enforcement capacity, which is expected to limit labour market flexibility.

¹³ Amendments to the Construction Law passed in 2008 authorize SLI to inspect relatives of owners of a construction enterprise. These amendments were justified by the fact that cases of illegal employment detected in the course of inspections had often been explained by family relationships between workers and owners of construction enterprises.

¹⁴ SLI data on 2010 is not yet available.

3.3 Tax burden on labour

The effect of labour taxation on the labour market operates through the wedge between the labour costs faced by employers and take-home pay received by employees. On the labour demand side, there are two effects of higher taxation on employment, both working in the same direction: first, higher labour costs induce employers to substitute relatively expensive labour for capital, and second – working through product price impact – higher labour costs may lead to higher product prices which reduces product demand and hence demand for labour. On the labour supply side, higher taxation triggers two effects, which work in opposite directions: the substitution effect implies that individuals substitute more expensive leisure to work (labour supply rises), while the income effect means that the demand for leisure increases (labour supply declines). The ultimate impact of a tax increase on employment and wages depends on relative elasticities of labour supply and demand, as well as on price elasticity of output.

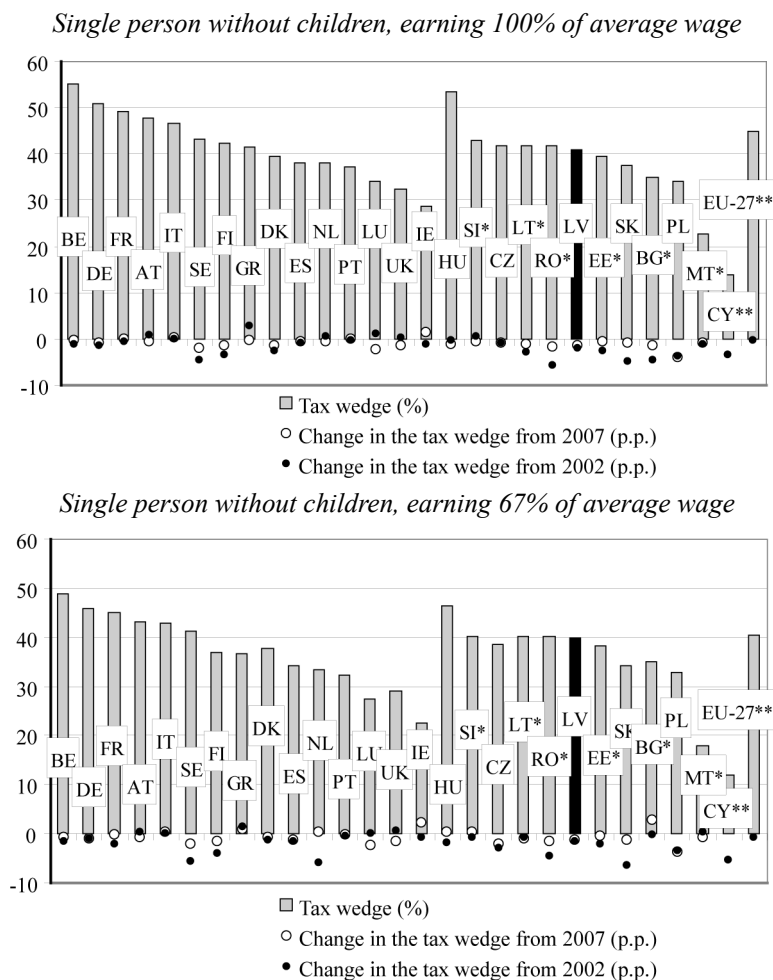
The impact of taxation on the labour market can also be affected by other labour market institutions. For example, wage setting institutions, such as trade unions or the minimum wage, can affect an employer's ability to shift the tax burden to employees. Daveri and Tabellini (1997) have shown that an increase in the tax wedge has a smaller impact on the unemployment rate in countries with decentralised wage bargaining. Also, any labour market institution which increases non-employment income, e.g., unemployment or other social benefits, by strengthening an employee's fallback position, increases labour supply elasticity and, therefore, amplifies the negative impact of the tax burden on employment (Gora et al, 2006).

The impact of taxation on employment also depends on the prevalence of informal economic activity (Behar, 2009). An increase in the tax burden can reduce employment and participation rate if employees respond to the increase by leaving formal employment. Alternatively, if employees choose to leave formal employment and register as unemployed, the increase in the tax burden leads to lower employment and higher unemployment without affecting the participation rate.

In this paper, two alternative indicators are used to analyse the Latvian tax burden on labour and to make cross-country comparisons – the tax wedge and the implicit tax rate. The tax wedge can be calculated for a variety of households of different composition with different levels of income. It reflects the difference between labour costs to employers and the net take-home pay of workers. This is a purely theoretical indicator, based on tax laws, and does not include actual tax revenue data. The implicit tax rate, on the other hand, is based on actual tax revenues and is calculated as a ratio of budget revenues from labour taxes (personal income tax and social security contributions in the case of Latvia) to total compensation of employees from the national accounts.

The difference between these two indicators can be substantial if the prevalence of so-called envelope wages is high. The tax wedge is not affected by unreported wages, since it is based on the amount of taxes that an individual is obliged to pay and is not affected by actual tax payments. The implicit tax rate is in turn influenced by actual tax revenues. Moreover, data on the denominator in the implicit tax rate formula, i.e., data on compensation of employees, come from national accounts. This means that those data are adjusted for the amount of unre-

Figure 9: Tax wedge on labour in EU member states in 2009 (%) and tax wedge change compared to 2002 and 2007 (percentage points)



* Data on 2008

** Data on 2007

Source: European Commission, 2010, author's calculations

Note: data on 2009 are provisional (European Commission, 2010), data on Latvia – author's calculations

ported wages by statistical offices. Therefore, if the prevalence of unreported wages is high, implicit tax may suggest a lower tax burden on labour than the tax wedge.

The tax burden on labour in “new” EU countries is generally quite high, not much lower than in the “old” member states. High labour taxes in the EU-12 are partly a consequence of the transition period (Anspal and Vörk, 2007). Output contraction in the initial stage of transition and a sharp increase in unemployment in these countries required substantial expenditure in the social sphere, which was financed via raising labour taxes.

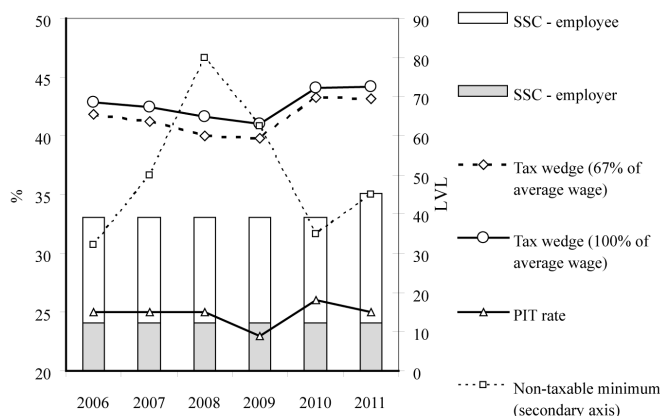
The tax wedge on labour in the “old” and “new” EU member states in 2009 is presented in Figure 9.

Although most EU-27 countries have reduced the tax wedge in recent years, it still remains high relative to developed countries outside the EU (e.g., the tax wedge for a single childless individual earning the average wage in 2009 in the USA was 29.4%, but for a single childless individual earning 67% of the average wage – 26.9%; in Australia – 26.7% and 20.7%, respectively (OECD, 2011)). In many EU-27 countries, despite the tough fiscal situation, taxes were significantly cut in 2008 and particularly in 2009, as many countries implemented measures aimed at reducing labour costs and stimulating their labour markets to alleviate the post-crisis adjustment. Thus, the personal income tax rate was significantly reduced in Denmark, Hungary, Finland and Sweden, while some other countries modified tax brackets or implemented other changes in their labour taxation system (Germany, Spain and Italy) (European Commission, 2010).

The tax wedge in Latvia in 2009 for an individual earning the average wage was below the EU-27 average (41.0% and 44.8%, respectively), but for an individual earning two thirds of the average wage – roughly at the EU-27 level (39.8% and 40.6%). The tax wedge was higher than in Estonia for both individual types and about the same as in Lithuania.

In contrast to many EU-15 countries, the tax wedge in Latvia was increased after 2008 (see Figure 10). Initially, in mid 2009, the government undertook an effort to shift the tax burden from labour to consumption, and reduced the personal income tax rate from 25% to 23%, but for self-employed (not shown on the graph) the rate was cut to 15%. Non-taxable income was cut, yet the tax wedge on the whole was reduced for both average-wage and low-wage earners

Figure 10: Rate of mandatory social security contributions (SSC) for employees and employers (%), personal income tax (PIT, %), tax exempt income (LVL, right-hand axis) and tax wedge for childless person earning 100% and 67% of the average wage (%) in Latvia in 2006-2011

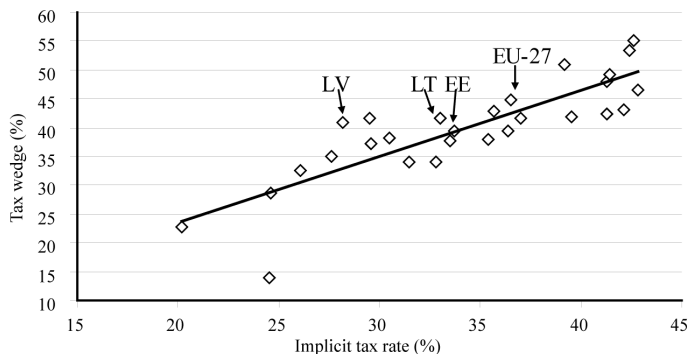


Source: Latvijas Vēstnesis (2011), author's calculations

(from 41.6% to 41.0% and from 40.0% to 39.8%, respectively). In 2010 and 2011, however, under pressure of the necessity to achieve fiscal consolidation, the government increased the personal income tax rate (to 26% in 2010 and then cut to 25% in 2011) and the rate of social security contributions paid by employees (from 9% to 11% in 2011). Moreover, non-taxable income was further reduced. As a result, the tax wedge on average wage earners increased to 44.2% in 2011, but on low wage earners – to 43.2%.

At the same time, the implicit tax rate suggests that the tax burden on labour in Latvia, compared to the EU-27 average, and Estonia and Lithuania, is lower¹⁵ (see Figure 11). There are certain limitations to comparing the implicit tax on labour with the tax wedge on labour, since the implicit tax rate does not allow accounting for progressivity of a tax system, and it can be affected by the demographic composition of the population. Nevertheless, it is noteworthy that the implicit tax rate on labour in Latvia is significantly below that in Lithuania and Estonia, countries both having flat personal income tax, while the tax wedge in Latvia is higher than in Estonia and only slightly below that in Lithuania. This result might be an indication of a relatively high incidence of unreported wages in Latvia.

Figure 11: Implicit tax rate on labour (2008, %) vs. tax wedge for a single childless person earning average wage (% , 2009*) in EU



* Data on Bulgaria, Estonia, Lithuania, Malta, Romania and Slovenia refer to 2008, data on Cyprus and EU-27 refer to 2007

Source: Eurostat, European Commission, 2010, Latvian Vēstnesis (2011), author's calculations

from the formal rules, being provoked by the strictness of the rules, undermine the merit of labour market institutions, which are there in the first place to protect workers and increase their welfare.

There is also evidence suggesting that tax payment procedures in Latvia are relatively complicated. According to World Bank data (World Bank et al, 2010), the average time that a Latvian entrepreneur spent on complying with their labour tax obligations in 2009 was 165 hours (there were only 4 countries in the EU where entrepreneurs spent more time on paying taxes – Finland (200 hours), Italy (214 hours), the Czech Republic (262 hours) and Bulgaria (288 hours)). This compares to entrepreneurs in Luxembourg who spent 14 hours, Estonia – 34 hours and Sweden – 36 hours, while the average time across the EU-27 was 108 hours.

¹⁵ Latest available data on implicit tax rate on labour income refers to 2008.

To conclude, formal tax legislation implies that the tax burden on labour in Latvia on average-wage and low-wage earners is approximately at the EU-27 level, it is higher than in Estonia and slightly lower than in Lithuania. Moreover, in the period after 2008 most EU-15 countries implemented measures aimed at reducing labour costs, while in Latvia the tax burden on labour was increased. Differences between the implicit tax rate on labour and the tax wedge on labour suggest that the prevalence of unreported wages in Latvia might be higher than in Estonia and Lithuania.

4. Conclusions

Analysis of the labour market institutions considered suggests that employment protection legislation and the tax burden on labour are not likely to impede Latvian labour market post-crisis adjustment. A relatively high minimum wage might be a factor hampering employment recovery, at least in the formal segment of the labour market.

However, there is evidence suggesting that the flexible reaction of the labour market is to a large extent achieved by employers circumventing the formal rules. This in itself creates distortions in the market - the efficiency of the economy could be increased if resources spent on getting around the regulations were spent in more productive ways. Moreover, deviations from the formal rules, being provoked by the strictness of the rules, undermine the merit of labour market institutions, which are there in the first place to protect workers and increase their welfare.

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